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DATA 318

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MIAC Baseball Predictor

For our DATA Mining Final Project, we decided to explore creating a MIAC baseball predictor. We are both Concordia baseball players, which makes this an interesting topic for us to dive into. Our target variable is binary classification outcome: whether a team wins or loses a MIAC game. Our goals in this project are to create a successful classification model for predicting the outcome to MIAC games, and learning what variables/statistics are the most important for predicting the outcome to baseball games.

Our primary data source will be the MIAC athletics website, which provides game results, box scores, and season statistics in a structured table format. We will scrape this data and convert it into CSV files for model analysis.

Our analysis plan includes several steps. First, we plan to clean and merge the data into a game-level dataset, and make engineering features like rolling average (last 5 games stats) and pitcher performance. We will then explore multiple classification models, including Logistic Regression, LASSO, and BeSS, chosen for their interpretability and ability to perform feature selection. We might also compare these to tree-based models like Random Forest to see whether more flexible models improve predictive accuracy.

Model performance will be evaluated using train/test splits, cross-validation, and metrics such as accuracy, ROC/AUC, and confusion matrices. Our goal is not only to build a model that

predicts MIAC baseball outcomes effectively, but also to identify which variables are most important in determining game success.

Bibliography

Minnesota Intercollegiate Athletic Conference. *2026 Baseball Cumulative Statistics*. MIAC Athletics, 2026, <https://miacathletics.com/stats.aspx?path=baseball&year=2026>.